THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. By Euclid's algorithm, gcd(84, 60) is: 1

(a) 6 **(b)** 24 **(c)** 12 **(d)** 13

2. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(50, 8) is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 12

3. Using the long-division (Āryabhaṭa) form of Euclid's algorithm, the number of reductions needed for gcd(825, 375) is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

4. The improved gcd algorithm that keeps most-recent-common-divisor scans k from 2 up to min(12, 30). The number of values of k scanned is: 1

(a) 12 **(b)** 29 **(c)** 11 **(d)** 10

5. The list of common divisors of 60 and 24, in increasing order, ends with: 1

(a) 60 **(b)** 13 **(c)** 6 **(d)** 12

6. In a parallelogram ABCD, $\angle A = 48°$. Then $\angle B$ is: 1

(a) 132° **(b)** 66° **(c)** 48° **(d)** 312°

7. In $\triangle ABC$, P and Q are the midpoints of AB and AC and BC = 14 cm. The length of PQ is: 1

(a) 28 cm **(b)** 14 cm **(c)** 9 cm **(d)** 7 cm

8. The median AD of $\triangle ABC$ has length 30 cm and G is the centroid. Then AG is: 1

(a) 20 cm **(b)** 15 cm **(c)** 10 cm **(d)** 28 cm

9. The sides of a triangle are 12 cm, 16 cm and 20 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: 1

(a) 46 cm **(b)** 24 cm **(c)** 26 cm **(d)** 48 cm

10. In a parallelogram ABCD, $\angle A = (4x + 10)°$ and $\angle B = (2x + 20)°$. The value of x is: 1

(a) 30 **(b)** 15 **(c)** 25 **(d)** 20

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write an algorithm prime(n) to test whether n is prime, using divisors(n). 2

12. The prime factorisation of 180 is $2^2 \times 3^2 \times 5^1$. Suggest a way to store it as a list and give the list. 2

13. The diagonals of parallelogram ABCD meet at O, with AC $= 14$ cm and BD $= 10$ cm. Find OA and OB. 2

14. Why must the angles meeting at a vertex of a tiling of the plane add up to $360°$? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. **Case study:** Adding two numbers by counting dots takes 1 second per dot. Ria adds 3-digit numbers (about 800 dots in all). 3

(i) How long, in minutes, for 800 dots?

(ii) About how many dots for 5-digit numbers (100 times as many)?

(iii) How many extra columns does the column method need for 5-digit numbers?

16. **Case study:** A triangular plot has medians of lengths 36 m, 27 m and 45 m. A water tank is placed at the centroid. **3**

(i) Find the distance of the centroid from the vertex on the 36 m median.

(ii) Find its distance from the midpoint of that side.

(iii) State the theorem that gives these.

17. **Case study:** Meera follows the algorithm: 'Start with count = 0. For each j from 1 to 12, if j divides 12, add 1 to count.' **3**

(i) What is the final value of count?

(ii) What does count represent?

(iii) Which numbers j add to count?

OR

Case study: Arjun runs Euclid's subtraction algorithm on gcd(7, 3) and Āryabhaṭa's algorithm on the same pair. **3**

(i) Write the chain of pairs for subtraction.

(ii) Write the chain for division.

(iii) Which is shorter and by how many reductions?

18. **Case study:** A floor is tiled with copies of the irregular quadrilateral SOME with angles 70°, 80°, 100°, 110°. **3**

(i) Find the sum of its angles.

(ii) Four copies meet at a point using each angle once. What is the total angle there?

(iii) Does this show that the plane can be tiled with it?

OR

Case study: A kite-shaped signboard ABCD has diagonals AC $= 24$ cm and BD $= 10$ cm meeting at right angles; E, F, G, H are the midpoints of AB, BC, CD, DA. **3**

(i) Name EFGH.

(ii) Find its sides.

(iii) Find its perimeter.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Neha writes the algorithm: 'Start with most-recent-common-divisor = 1. For each k from 2 to min(m, n): if k divides both m and n, update to k.' **5**

(i) Execute it for m = 20, n = 30 and list the k that update the value.

(ii) What is the final value and what does it equal?

(iii) How many values of k were examined?

(iv) Suggest how to reduce the number of checks using Euclid's idea.

OR

Case study: The chapter says the number of reductions in Euclid's subtraction algorithm for gcd(2k + 1, 2) is about k. **5**

(i) For k = 5, write the pair-chain for gcd(11, 2) and count the reductions.

(ii) Write the division-algorithm chain.

(iii) Compare.

(iv) What does this show about the efficiency of the two algorithms?

20. **Case study:** A tiler wants to tile a floor with identical copies of a non-convex quadrilateral DART with angles 40°, 50°, 30° and 240°. **5**

(i) Check the angle sum.

(ii) Four copies are placed around a point using each angle once; what is the angle total?

(iii) Describe Method 1 for adding more tiles.

(iv) Why can the plane not be tiled with regular pentagons?

OR

ABCD is a parallelogram with $AB \neq BC$. Prove that the four angle bisectors meet pairwise at the vertices of a rectangle. Why is $AB \neq BC$ assumed? **5**

END OF QUESTION PAPER — Total: 40 Marks

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